# PROJECT PHASE-3

**Demon Slayer Stock Market System**

**1. ER to Relational Model Conversion**

## Mapping Approach

• **Strong Entities**:Mapped directly to relations
(SECTOR, INVESTOR, COMPANY, PRODUCT, STOCK, FIGHT, MAGIC_SPELL, NEWS_EFFECT, TRANSACTION)

• **Weak Entities**:Included owner's primary key as part of the composite primary key
(INVESTOR_SKILLS, FIGHT_SCORE)

• **Subclasses (Specialization):**Created separate relations for subclasses
HASHIRA(InvestorID PK/FK)
UPPER_RANK_DEMON(InvestorID PK/FK)

• **Many-to-Many Relationships**:Created junction tables
(PORTFOLIO, PROMOTES, CASTS_SPELL_ON)

• **Recursive Relationships**:MENTORSHIP implemented using MentorID as FK in INVESTOR

• **Multivalued Attributes:**Separated into independent relations
HASHIRA_BREATHING(InvestorID, BreathingTechnique)
COMPANY_PRODUCT_TYPE(CompanyID, ProductType)

• **Ternary & Quaternary Relationships**:Mapped to separate relations with composite keys
(CASTS_SPELL_ON, INVESTOR_PRODUCT_FIGHT_REWARD, INVESTOR_FIGHT_SKILL_SCORE)

**Changes made:**

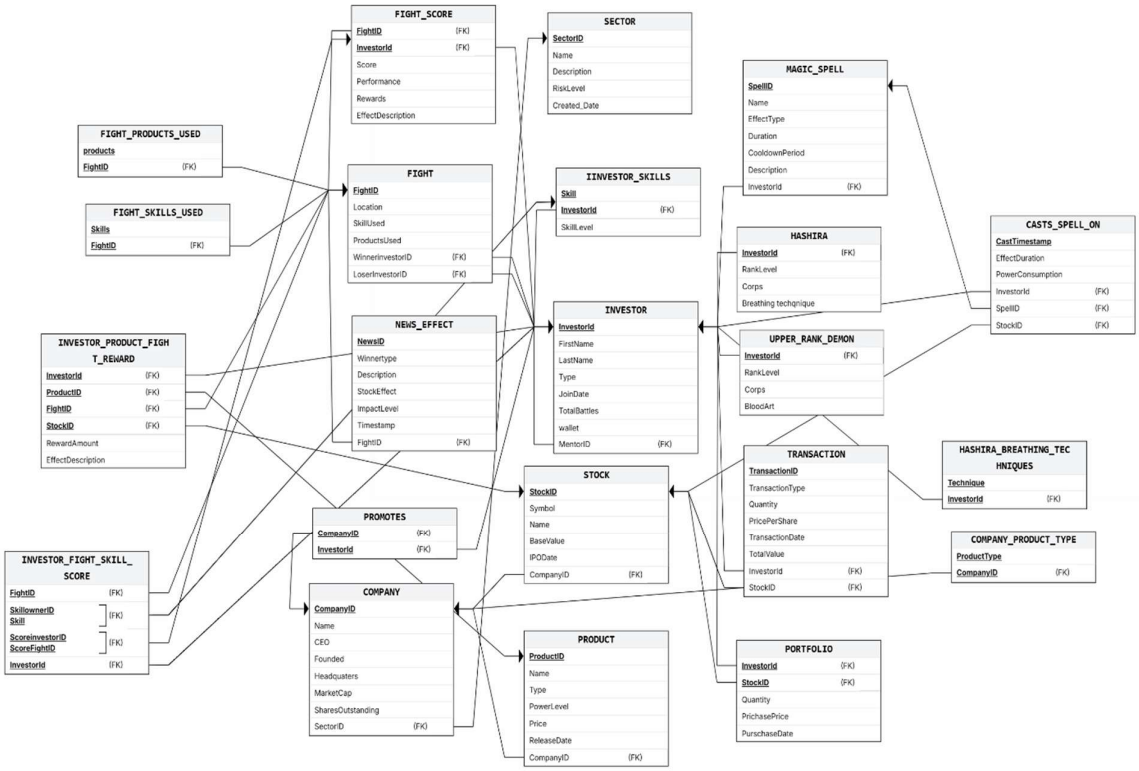
**-> Multivalued attributes split into separate tables**

- HASHIRA_BREATHING
- COMPANY_PRODUCT_TYPE
- FIGHTS_PRODUCT_USED
- FIGHTS_SKILL_USED

**->Removed derived attributes:**

- WinRate
- TotalValue
- ShareValue

**Relational Schema After ER Mapping**




## First Normal Form (1NF) Conversion

**Analysis:**

- Multivalued attributes
- Non-atomic (composite or repeating) attributes
- Attributes storing lists or grouped values

**Findings:**

- All multivalued attributes from the ER diagram (e.g., BreathingTechnique, ProductType) were already mapped to separate relations during ER-to-relational conversion.
- Non-atomic fields in FIGHT (SkillsUsed, ProductsUsed) were removed and represented through FIGHT_SKILLS_USED and FIGHT_PRODUCTS_USED relations.
- No table contains repeating groups or multi-valued cells after mapping.
- Every attribute across all relations contains atomic, single-valued data.

**Conclusion:**

The relational schema already satisfies 1NF requirements. No additional changes were required to achieve First Normal Form.

**“””NO MODIFICATIONS NEEDED FOR 1NF.”””**

## Second Normal Form (2NF) Analysis

**Analysis:**Examined all tables with composite primary keys for partial dependencies:

- INVESTOR_SKILLS: SkillLevel depends on full key (InvestorID, Skill)
- FIGHT_SCORE: All attributes depend on full key (FightID, InvestorID)
- PORTFOLIO: All attributes depend on full key (InvestorID, StockID)
- CASTS_SPELL_ON: EffectDuration and PowerConsumption vary per casting instance, depend on full key
- All other composite-key tables: No non-key attributes or no partial dependencies

**Conclusion:**

**“””NO MODIFICATIONS NEEDED. All 20 tables already satisfy 2NF requirements**.”””

## Third Normal Form (3NF) Conversion

**hird Normal Form (3NF) Analysis**

**Analysis:**

• All non-key attributes must depend only on the primary key.
• Derived attributes must be removed because they introduce transitive dependency and redundancy.

**Findings:**

• **INVESTOR:**
**–** *WinRate* is a derived attribute (calculated from wins and TotalBattles).
– This introduces a transitive dependency on non-key attributes.
– WinRate removed to satisfy 3NF.

• **STOCK:**
**–** *ShareValue* is derived from BaseValue and market changes.
– Not functionally dependent solely on StockID.
– ShareValue removed.

• **TRANSACTION:**
**–** *TotalValue* = Quantity × PricePerShare, hence derived.
– Storing it creates redundancy and update anomalies.
– TotalValue removed.

• **COMPANY:**
**–** *MarketCap* is retained because it is not derived from stored attributes in this schema and depends on external market factors.
– No transitive dependency identified**.**

• **All other relations:**
**–** No non-key attributes that transitively depend on other non-key attributes.
– Already satisfied 3NF.

**Conclusion:**

Derived attributes (WinRate, ShareValue, TotalValue) were removed to eliminate transitive dependencies. After these corrections, all relations satisfy Third Normal Form, and no further changes are required. **“”””NO MODIFICATIONS NEEDED for 3NF requirements**.”””””

**Relational Schema After 3NF,**

